MEDIUM · ADVANCED MATH · ANSWER KEY

Advanced Math

03

10 answers · Rationale-rich · Teach from doc

Q1

D

D. $x^{-2}y^3z^{10} + x^{-6}y^{11}z^{-2}$

Choice D is correct. The product of $x^{-6}y^3z^5$ and $x^4z^5 + y^8z^{-7}$ can be represented by the expression $x^{-6}y^3z^5x^4z^5 + y^8z^{-7}$. Applying the distributive property to this expression yields $x^{-6}y^3z^5x^4z^5 + x^{-6}y^3z^5y^8z^{-7}$, or $x^{-6}x^4y^3z^5z^5 + x^{-6}y^3y^8z^5z^{-7}$. This expression is equivalent to $x^{-6+4}y^3z^{5+5} + x^{-6}y^{3+8}z^{5-7}$, or $x^{-2}y^3z^{10} + x^{-6}y^{11}z^{-2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q2

410

The correct answer is 410. It's given that t minutes after an initial observation, the number of bacteria in a population is $60{,}0002^{\frac{t}{410}}$. This expression consists of the initial number of bacteria, 60,000, multiplied by the expression $2^{\frac{t}{410}}$. The time it takes for the number of bacteria to double is the increase in the value of t that causes the expression $2^{\frac{t}{410}}$ to double. Since the base of the expression $2^{\frac{t}{410}}$ is 2, the expression $2^{\frac{t}{410}}$ will double when the exponent increases by 1. Since the exponent of the expression $2^{\frac{t}{410}}$ is $\frac{t}{410}$, the exponent will increase by 1 when t increases by 410. Therefore the time, in minutes, it takes for the number of bacteria in the population to double is 410.

Q3

5

The correct answer is 5. Multiplying both sides of the given equation by $x + 6$ results in $55 = xx + 6$. Applying the distributive property of multiplication to the right-hand side of this equation results in $55 = x^2 + 6x$. Subtracting 55 from both sides of this equation results in $0 = x^2 + 6x - 55$. The right-hand side of this equation can be rewritten by factoring. The two values that multiply to -55 and add to 6 are 11 and -5. It follows that the equation $0 = x^2 + 6x - 55$ can be rewritten as $0 = x + 11x - 5$. Setting each factor equal to 0 yields two equations: $x + 11 = 0$ and $x - 5 = 0$. Subtracting 11 from both sides of the equation $x + 11 = 0$ results in $x = -11$. Adding 5 to both sides of the equation $x - 5 = 0$ results in $x = 5$. Therefore, the positive solution to the given equation is 5.

Q4

B

B. $x^4 + 23x^2 + 96$

Choice B is correct. The expression $x^2 + 11^2$ can be written as $x^2 + 11x^2 + 11$, which is equivalent to $x^2x^2 + 11 + 11x^2 + 11$. Distributing x^2 and 11 to $x^2 + 11$ yields $x^4 + 11x^2 + 11x^2 + 121$, or $x^4 + 22x^2 + 121$. The expression $x - 5x + 5$ is equivalent to $x - 5x + x - 55$. Distributing x and 5 to $x - 5$ yields $x^2 - 5x + 5x - 25$, or $x^2 - 25$. Therefore, the expression $x^2 + 11^2 + x - 5x + 5$ is equivalent to $x^4 + 22x^2 + 121 + x^2 - 25$, or $x^4 + 22x^2 + 121 + x^2 - 25$. Combining like terms in this expression yields $x^4 + 23x^2 + 96$.

Choice A is incorrect. Equivalent expressions must be equivalent for any value of x. Substituting 0 for x in this expression yields -14, whereas substituting 0 for x in the given expression yields 96.

Choice C is incorrect. Equivalent expressions must be equivalent for any value of x. Substituting 0 for x in this expression yields 121, whereas substituting 0 for x in the given expression yields 96.

Choice D is incorrect. Equivalent expressions must be equivalent for any value of x. Substituting 0 for x in this expression yields 146, whereas substituting 0 for x in the given expression yields 96.

Q5

B

B. $C = 600(0.85)^t$

Choice B is correct. A model for a quantity C that decreases by a certain percentage per time period t is an exponential equation in the form $C = I\left(1 - \frac{r}{100}\right)^t$, where I is the initial value at time $t = 0$ for r% annual decline. It's given that C is the number of millions of CDs sold in the United States and that t is the number of years after 2005. It's also given that 600 million CDs were sold at time $t = 0$, so $I = 600$. This number declines by 15% per year, so $r = 15$. Substituting these values into the equation produces $C = 600\left(1 - \frac{15}{100}\right)^t$, or $C = 600(0.85)^t$.

Choice A is incorrect and may result from errors made when representing the percent decline. Choices C and D are incorrect. These equations model exponential increases in CD sales, not exponential decreases.

Q6

D

D. $T = \frac{v - 331.3}{0.606}$

Choice D is correct. To express T in terms of v, subtract 331.3 from both sides of the equation, which gives v – 331.3 = 0.606T. Dividing both sides of the equation by 0.606 gives $\frac{v - 331.3}{0.606} = T$.

Choices A, B, and C are incorrect and are the result of incorrect steps while solving for T.

Q7

B

B. $1 - p^7$

Choice B is correct. Multiplying $(1 - p)$ by each term of the polynomial within the second pair of parentheses gives $(1 - p)1 = 1 - p$; $(1 - p)p = p - p^2$; $(1 - p)p^2 = p^2 - p^3$; $(1 - p)p^3 = p^3 - p^4$; $(1 - p)p^4 = p^4 - p^5$; $(1 - p)p^5 = p^5 - p^6$; and $(1 - p)p^6 = p^6 - p^7$. Adding these seven expressions together and combining like terms gives $1 + (p - p) + (p^2 - p^2) + (p^3 - p^3) + (p^4 - p^4) + (p^5 - p^5) + (p^6 - p^6) - p^7$, which can be simplified to $1 - p^7$.

Choices A, C, and D are incorrect and may result from incorrectly identifying the highest power of p in the expressions or incorrectly combining like terms.

Q8

B

B. The value of the bank account is estimated to be approximately 243 dollars in 1962.

Choice B is correct. It's given that the function $f x = 206 1.034^x$ models the value, in dollars, of a certain bank account by the end of each year from 1957 through 1972, where x is the number of years after 1957. It follows that $f x$ represents the estimated value, in dollars, of the bank account x years after 1957. Since the value of $f 5$ is the value of $f x$ when $x = 5$, it follows that "$f 5$ is approximately equal to 243" means that $f x$ is approximately equal to 243 when $x = 5$. In the given context, this means that the value of the bank account is estimated to be approximately 243 dollars 5 years after 1957. Therefore, the best interpretation of the statement "$f 5$ is approximately equal to 243" in this context is the value of the bank account is estimated to be approximately 243 dollars in 1962.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q9 B

B. -4

Choice B is correct. Adding y to each side of the second equation in the given system of equations yields $8x + 16 = y$. Substituting $8x + 16$ for y in the first equation yields $x^2 + 8x + 16 + 10 = 10$. Subtracting 10 from each side of this equation yields $x^2 + 8x + 16 = 0$. This equation can be rewritten as $x + 4^2 = 0$. Taking the square root of each side of this equation yields $x + 4 = 0$. Subtracting 4 from each side of this equation yields $x =$ -4. Therefore, the value of x is -4.

Choice A is incorrect. This is the value of y, not x.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10 A

A. $-\frac{1}{6}x+8$

Choice A is correct. By distributing the minus sign through the expression $\left(\frac{2}{3}x-5\right)$, the given expression can be rewritten as $\left(\frac{1}{2}x+3\right)-\frac{2}{3}x+5$, which is equivalent to $\frac{1}{2}x-\frac{2}{3}x+3+5$. Combining like terms gives $\left(\frac{1}{2}-\frac{2}{3}\right)x+(3+5)$, or $-\frac{1}{6}x+8$.

Choice B is incorrect and may be the result of failing to distribute the minus sign appropriately through the second term and simplifying the expression $\frac{1}{2}x+3-\frac{2}{3}x-5$. Choice C is incorrect and may be the result of multiplying the expressions $\left(\frac{1}{2}x+3\right)$ and $\left(-\frac{2}{3}x+5\right)$. Choice D is incorrect and may be the result of multiplying the expressions $\left(\frac{1}{2}x+3\right)$ and $\left(-\frac{2}{3}x-5\right)$.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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